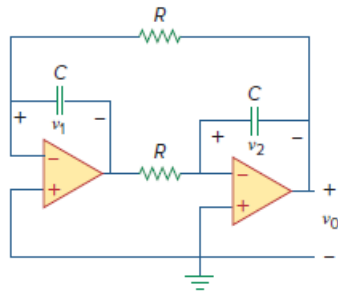


Problem

Given the op amp circuit in Fig. 16.90, if $v_1(0^+) = 2 \text{ V}$ and $v_2(0^+) = 0 \text{ V}$, find v_0 for $t > 0$. Let $R = 100 \text{ k}\Omega$ and $C = 1 \mu\text{F}$.

Figure 16.90:



Step-by-step solution

Step 1 of 6

Refer to Figure 16.90 from the textbook.

Consider the following values for the components:

$$R = 100 \text{ k}\Omega$$

$$C = 1 \mu\text{F}$$

Consider the following values for the initial voltages across the capacitors:

$$v_1(0^+) = 2 \text{ V}$$

$$v_2(0^+) = 0 \text{ V}$$

Represent the circuit in s-domain including the initial conditions.

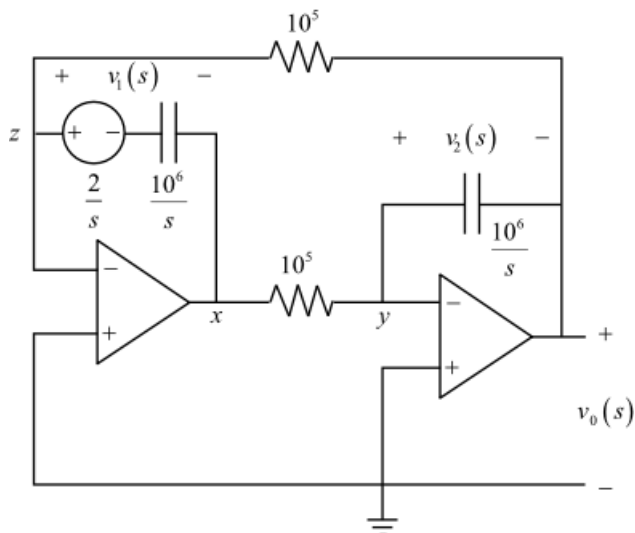


Figure 1

Step 2 of 6

Assume that the voltages at the nodes x , y , and z are $V_x(s)$, $V_y(s)$, and $V_z(s)$ respectively.

Consider that the voltages at the input terminals of the operational amplifier are equal, due to the virtual ground property of operational amplifier. Therefore, the voltages at the nodes y and z are equal to zero, since the positive terminals of both the operational amplifiers are grounded.

Apply Kirchhoff's current law at node x .

$$\begin{aligned} \frac{V_x(s) - V_y(s)}{10^5} + \frac{V_x(s) - [V_z(s) - v_1(0^+)]}{\frac{10^6}{s}} &= 0 \\ \frac{V_x(s) - V_y(s)}{10^5} + \frac{s \{V_x(s) - [V_z(s) - v_1(0^+)]\}}{10^6} &= 0 \\ V_x(s) \left[\frac{1}{10^5} + \frac{s}{10^6} \right] - V_y(s) \left[\frac{1}{10^5} \right] - [V_z(s) - v_1(0^+)] \left[\frac{s}{10^6} \right] &= 0 \\ V_x(s) \left[\frac{1}{10^5} + \frac{s}{10^6} \right] = V_y(s) \left[\frac{1}{10^5} \right] + [V_z(s) - v_1(0^+)] \left[\frac{s}{10^6} \right] \end{aligned}$$

Step 3 of 6

Substitute 0 for $V_y(s)$ and $V_z(s)$ in the expression:

$$\begin{aligned} V_x(s) \left[\frac{1}{10^5} + \frac{s}{10^6} \right] &= (0) \left[\frac{1}{10^5} \right] + [0 - v_1(0^+)] \left[\frac{s}{10^6} \right] \\ &= -v_1(0^+) \left[\frac{s}{10^6} \right] \end{aligned}$$

Substitute $\frac{2}{s}$ for $v_1(0^+)$.

$$\begin{aligned} V_x(s) \left[\frac{1}{10^5} + \frac{s}{10^6} \right] &= -\left(\frac{2}{s} \right) \left(\frac{s}{10^6} \right) \\ V_x(s) &= \frac{-\left(\frac{2}{10^6} \right)}{\frac{1}{10^5} + \frac{s}{10^6}} \\ &= \frac{-\left(\frac{2}{10^6} \right)}{\left(\frac{10 + s}{10^6} \right)} \\ V_x(s) &= \frac{-2}{s + 10} \dots\dots (1) \end{aligned}$$

Step 4 of 6

Similarly apply Kirchhoff's current law at node y .

$$\frac{V_y(s) - v_0(s)}{\frac{10^6}{s}} + \frac{V_y(s) - V_x(s)}{10^5} = 0$$

$$V_y(s) \left[\frac{s}{10^6} + \frac{1}{10^5} \right] - \frac{V_x(s)}{10^5} - v_0(s) \left[\frac{s}{10^6} \right] = 0$$

$$\frac{V_x(s)}{10^5} + v_0(s) \left[\frac{s}{10^6} \right] = V_y(s) \left[\frac{1}{10^5} + \frac{s}{10^6} \right]$$

Substitute 0 for $V_y(s)$ in the expression:

$$\frac{V_x(s)}{10^5} + v_0(s) \left[\frac{s}{10^6} \right] = (0) \left[\frac{1}{10^5} + \frac{s}{10^6} \right]$$

$$\frac{V_x(s)}{10^5} + v_0(s) \left[\frac{s}{10^6} \right] = 0$$

$$v_0(s) \left[\frac{s}{10^6} \right] = -\frac{V_x(s)}{10^5}$$

$$v_0(s) = -\frac{\frac{V_x(s)}{10^5}}{\frac{s}{10^6}}$$

$$= -V_x(s) \left[\frac{10}{s} \right]$$

Step 5 of 6

Substitute the expression for $V_x(s)$ from equation (1).

$$v_0(s) = -\left(\frac{-2}{s+10} \right) \left(\frac{10}{s} \right)$$

$$= \frac{20}{s(s+10)}$$

Use partial fraction method to simplify the function further.

$$\frac{20}{s(s+10)} = \frac{A}{s} + \frac{B}{s+10}$$

$$20 = A(s+10) + Bs$$

$$= As + 10A + Bs$$

$$= (A+B)s + 10A$$

Compare the coefficients of the terms to get the following values:

$$10A = 20$$

$$A = 2$$

$$A + B = 0$$

$$2 + B = 0$$

$$B = -2$$

Write the simplified expression for $v_0(s)$.

$$v_0(s) = \frac{2}{s} - \frac{2}{s+10}$$

$$= \frac{2}{s} - \frac{2}{s+10}$$

Step 6 of 6

Find the inverse Laplace transform to find the output voltage $v_0(t)$.

Use the following property of the Laplace transform:

$$u(t) \leftrightarrow \frac{1}{s}$$
$$e^{-at} \leftrightarrow \frac{1}{s+a}$$

Find the inverse Laplace transform of the function $v_0(s)$.

$$\begin{aligned} v_0(t) &= 2u(t) - 2e^{-10t}u(t) \\ &= 2[1 - e^{-10t}]u(t) \end{aligned}$$

Thus, the output voltage $v_0(t)$ for $t > 0$ is $\boxed{v_0(t) = 2[1 - e^{-10t}]u(t) \text{ V}}$.